

Coherent Prime-Square Blocks on Literal Residual Fibers: Exact Gram Geometry, Sharp Saturation, and a Route-Switch Certificate

Liang Wang

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Abstract

In the literal fixed-shift residual carrier, the terminal-prime coordinate is injective before it is forgotten, while the physical output coherently adds all terminal primes in one residual fiber. We isolate one actual fiber $y = (\gamma, s, j)$, write its inherited amplitudes as $b_{y,p}$, and study the prime square

$$\mathcal{G}_y = \left| \sum_{p \in \mathcal{P}_y} b_{y,p} \right|^2$$

against the resolved diagonal $\mathcal{D}_y = \sum_p |b_{y,p}|^2$. The exact Gram operator is the rank-one all-ones matrix. For fixed moduli $a_{y,p} = |b_{y,p}|$, the complete free-phase range is

$$[(2a_{y,\max} - R_y)_+^2, R_y^2], \quad R_y = \sum_p a_{y,p}.$$

Consequently the usual Bessel bound $\mathcal{G}_y \leq |\mathcal{P}_y| \mathcal{D}_y$ is sharp, and no coefficient-blind improvement follows from terminal-prime injectivity, support size, occupancy, or the diagonal energy. Equal moduli already admit both perfect saturation and maximal alignment on the same resolved support.

We give an exact coefficient-specific route certificate in terms of the physical alignment ratio, and a dominant-atom lower certificate. Applied to the literal TPC residual map, these are L1 identities and stop/go tests. They do not estimate the actual growing coefficients and therefore provide no L2 fixed-shift arithmetic saving, parity breakthrough, prime-pair asymptotic, or twin-prime conclusion.

Keywords: coherent prime square; residual fiber; rank-one Gram matrix; phase saturation; fixed shift; parity barrier.

1 Question and claim boundary

TPC-MVP1 lists a coherent prime-square test as an alternative attack on the generic signed gate [1]. The phrase “prime square” here means the squared norm of a coherent sum over terminal primes, not a restriction to integers p^2 .

The literal residual map of TPC-55 first keeps the terminal-prime label and then forgets it [3]. At one prescribed $h_0 \neq 0$, its physical output fiber has key

$$y = (\gamma, s, j), \quad \mathcal{P}_y = \{p : (\gamma, s, j, p) \text{ is active}\}.$$

The terminal-prime coordinate is injective before the forgetful map. Thus every $p \in \mathcal{P}_y$ has one inherited complex amplitude $b_{y,p}$, but all of them are coherently added at the output. The static-mask and resolved-energy results upstream do not estimate this final coherent sum [2, 3].

This paper answers the coefficient-blind part of the question completely. It does not claim a bound for the literal amplitudes as $X \rightarrow \infty$.

2 The literal block and its exact Gram operator

Definition 2.1 (Resolved and coherent energies). For a nonempty finite fiber $\mathcal{P}_y$, put

$$k_y = |\mathcal{P}_y|, \quad \mathcal{D}_y = \sum_{p \in \mathcal{P}_y} |b_{y,p}|^2, \quad \mathcal{G}_y = \left| \sum_{p \in \mathcal{P}_y} b_{y,p} \right|^2.$$

Also write

$$a_{y,p} = |b_{y,p}|, \quad R_y = \sum_p a_{y,p}, \quad a_{y,\max} = \max_p a_{y,p},$$

and

$$d_y = (2a_{y,\max} - R_y)_+.$$

The zero fiber is treated separately, with all quantities equal to zero.

Proposition 2.2 (Exact coherent-prime Gram identity). Order $\mathcal{P}_y = \{p_1, \dots, p_{k_y}\}$, put $b_y = (b_{y,p_i})_i$, and let $\mathbf{J}_{k_y} = \mathbf{1}\mathbf{1}^*$. Then

$$\boxed{\mathcal{G}_y = b_y^* \mathbf{J}_{k_y} b_y.}$$

The spectrum of $\mathbf{J}_{k_y}$ is $\{k_y, 0, \dots, 0\}$. Hence

$$0 \leq \mathcal{G}_y \leq k_y \mathcal{D}_y,$$

and the upper constant k_y is optimal.

Proof. The quadratic form expands to

$$b_y^* \mathbf{1}\mathbf{1}^* b_y = |\mathbf{1}^* b_y|^2 = \left| \sum_p b_{y,p} \right|^2.$$

The spectral statement is immediate. Cauchy–Schwarz gives the upper bound, and equality holds when all entries are equal. $\square$

Remark 2.3 (Where the obstruction sits). Prime-resolved injectivity controls the map before $\mathbf{J}_k$ is applied. It does not diagonalize $\mathbf{J}_k$. Distinct terminal primes are precisely the off-diagonal modes inside this rank-one quadratic form.

3 Sharp phase saturation

Theorem 3.1 (Complete free-phase interval). Fix a fiber and its moduli $(a_p)_{p \in \mathcal{P}}$. Allow the phases of $b_p = a_p e^{i\theta_p}$ to vary independently. Then

$$\boxed{\left\{ \left| \sum_p b_p \right|^2 : \theta_p \in \mathbb{R} \right\} = [d^2, R^2],} \quad d = (2a_{\max} - R)_+.$$

Every point of the interval is attained.

Proof. The possible lengths of a polygonal sum of segments of prescribed lengths a_p form the interval $[d, R]$. The lower endpoint is the reverse triangle inequality, attained by placing a longest segment opposite all others; when the longest segment does not dominate, the segments close to a polygon and the lower endpoint is zero. Continuous rotation of one segment, together with induction on the number of segments, fills the whole interval. Squaring gives the result. $\square$

Corollary 3.2 (Sharp saturation dichotomy). *The moduli alone force a positive coherent square if and only if*

$$a_{\max} > \frac{R}{2}.$$

In that case every phase assignment satisfies

$$\mathcal{G} \geq (2a_{\max} - R)^2.$$

If $a_{\max} \leq R/2$, the strongest universal modulus-only lower bound is zero.

Corollary 3.3 (Same-support nonidentifiability). *For every $k \geq 2$, there are two coefficient vectors on the same k -prime support with the same unit moduli and the same diagonal energy k , but with coherent squares 0 and k^2 , respectively.*

Proof. Use the k -th roots of unity in the first vector and k copies of 1 in the second. $\square$

Remark 3.4 (Physical versus comparison phases). The lower bound d_y^2 applies to the literal fiber because it uses its actual moduli. Attainability in [theorem 3.1](#) belongs to the larger free-phase comparison class. It does not say that the arithmetic coefficient realizes a root-of-unity phase pattern.

4 Coefficient-specific certificates

The finite geometry identifies exactly which additional statistic a coherent-prime argument must control.

Definition 4.1 (Alignment and cancellation ratios). For $\mathcal{D}_y > 0$, set

$$\alpha_y = \frac{\mathcal{G}_y}{k_y \mathcal{D}_y} \in [0, 1], \quad \beta_y = \frac{\mathcal{G}_y}{R_y^2} \in [0, 1].$$

For a finite family $\mathcal{B}$ of disjoint fibers, define

$$\mathcal{D}_{\mathcal{B}} = \sum_{y \in \mathcal{B}} \mathcal{D}_y, \quad \mathcal{G}_{\mathcal{B}} = \sum_{y \in \mathcal{B}} \mathcal{G}_y, \quad K_{\mathcal{B}} = \max_{y \in \mathcal{B}} k_y.$$

Proposition 4.2 (Exact block go certificate). *For every nonzero block,*

$$\frac{\mathcal{G}_{\mathcal{B}}}{K_{\mathcal{B}} \mathcal{D}_{\mathcal{B}}} = \sum_{y \in \mathcal{B}} \frac{k_y \mathcal{D}_y}{K_{\mathcal{B}} \mathcal{D}_{\mathcal{B}}} \alpha_y.$$

In particular, a coefficient-specific improvement

$$\mathcal{G}_{\mathcal{B}} \leq \varepsilon_X K_{\mathcal{B}} \mathcal{D}_{\mathcal{B}}$$

is equivalent to the displayed weighted alignment average being at most ε_X .

Proof. Substitute $\mathcal{G}_y = k_y \mathcal{D}_y \alpha_y$ and sum. $\square$

Proposition 4.3 (Dominant-prime lower certificate). *Let $\mathcal{B}_\epsilon$ be the fibers satisfying*

$$a_{y,\max} \geq \left(\frac{1}{2} + \epsilon\right) R_y.$$

Then

$$\boxed{\mathcal{G}_{\mathcal{B}} \geq 4\epsilon^2 \sum_{y \in \mathcal{B}_\epsilon} R_y^2.}$$

This bound is phase-independent and sharp at the single-fiber level.

Proof. On such a fiber, $d_y = 2a_{y,\max} - R_y \geq 2\epsilon R_y$. Apply [theorem 3.1](#) and sum. $\square$

Remark 4.4 (Upper saving and lower survival are different tasks). A dispersion route may seek small α_y , while an energy-survival route may seek a positive dominant share. The same unsigned occupancy statement cannot supply both. The two directions must be labeled separately.

5 Application to the TPC-109 queue item

On the literal TPC-55 carrier, the amplitudes $b_{y,p}$ include the source coefficient, masks, terminal-prime weight, fixed h_0 , and all retained profile factors. Therefore [propositions 2.2](#) to [4.3](#) and [theorem 3.1](#) apply without changing the physical output map. This is the positive content of the test.

The test also gives a rigorous stop certificate:

Theorem 5.1 (TPC-109 route decision). *Terminal-prime injectivity, the exact active prime support, subpower fiber multiplicity, and the resolved diagonal energy do not imply any coefficient-blind improvement over*

$$\mathcal{G}_{\mathcal{B}} \leq K_{\mathcal{B}} \mathcal{D}_{\mathcal{B}}.$$

Nor do they imply a positive lower comparison $\mathcal{G}_{\mathcal{B}} \gg \mathcal{D}_{\mathcal{B}}$. Any improvement in either direction must use one of:

- (i) *the literal phase relation among the actual $b_{y,p}$;*
- (ii) *a proved dominant-prime share;*
- (iii) *an arithmetic average that is subsequently localized back to the same prescribed h_0 ; or*
- (iv) *cancellation across a larger physical reassembly that retains all provenance.*

Proof. The upper bound and its sharpness are [proposition 2.2](#). The lower obstruction is [corollary 3.3](#). These examples retain support, moduli, diagonal energy, and multiplicity. Hence none of those unsigned invariants can decide the coherent square. $\square$

6 Level ledger and next gate

Statement	Status
Rank-one Gram identity and sharp phase interval	Unconditional finite-dimensional theorem (L0).
Identification with one literal residual fiber	Exact interface at prescribed h_0 (L1).
Coefficient-blind Bessel improvement	Rigorously stopped by same-support saturation.
Dominant-prime or alignment estimate for the growing coefficient	Not proved; arithmetic input required.
Fixed- h_0 polynomial saving, parity breach, or prime-pair theorem	Not claimed (L2 remains open).

The coherent-prime corridor therefore remains open only as a *coefficient-specific* route. TPC-110 should not recycle the resolved diagonal as if it were the post-bin determinant energy. It must analyze the exact binning kernel and the angle of the literal coefficient to that kernel.

Explicit exclusions. Nothing in this paper proves the generic affine Möbius hypothesis of TPC-MVP1, the determinant-energy bound, the distinguished zero-mode bound, the compatibility $\lambda_D \leq 2\eta_Z$, the independent physical endpoint $\Lambda_{\text{phys}} < 1/400$, a Hardy–Littlewood asymptotic, or the twin-prime conjecture. No statement is specialized to $h_0 = 2$.

References

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